



A novel deep wavelet convolutional neural network for actual ECG signal denoising

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ABSTRACT

Recently, more than 80% of sudden cardiac death is caused by arrhythmia, whose incidence has increased rapidly. In the actual wearable device acquisition process, ECG recordings are polluted by many different types of noise, mainly including baseline drift (BW), muscle artifact (MA) and electrode motion (EM). Therefore, the effective de-noising of ECG signals plays an important role in the subsequent accurate arrhythmia diagnosis. Nowadays, traditional filtering methods often only focus on one or two kinds of noise. When encountering different noises, different filtering algorithms need to be used, resulting in insufficient generalization ability of the algorithms. Aiming to effectively denoising ECG signals, this paper proposes a novel signal denoising method based on deep wavelet convolutional neural network. Inspired by the structure of the de-noising self-encoder, the convolution layers are used to replace the simple full-connected layers to build the encoder and decoder. Based on automatic feature extraction in convolution layer, the discrete wavelet transform is used to convert the signal into high-frequency and low-frequency components for replacing pooling layers to compress the input data and fully preserve the effective information. Finally, based on the dataset under the real noise conditions, compared with the existing methods, the proposed denoising model has better denoising performance, which reduces 0.194 RMSE and improves 5.99 SNR on average. Thus, the proposed denoising model has good potential in wearable devices.

1. Introduction

World Health Organization [1] presents that 31% of global deaths result from cardiovascular diseases, and arrhythmia is a common type of cardiovascular diseases. Electrocardiogram (ECG) is a convenient and non-invasive method to record cardiac electrical activity, which is used for detecting arrhythmia. With the development of wearable techniques, wearable devices provide the basis for more convenient and real-time long-range personal health monitoring, which is also the current trend in the development of the Internet of Things (IoT) [2]. For ECG monitoring systems, obtaining de-noising ECG signals plays an important role in subsequent automatic diagnosis. An ECG recording polluted by noise will have a great negative impact on the subsequent ECG analysis and greatly affect the diagnostic accuracy. Thus, accurate ECG signal denoising methods is of great significance to wearable sensors and devices.

In the actual wearable device acquisition process, ECG signals are polluted by many different types of noise, mainly including baseline drift (BW), muscle artifact (MA) and electrode motion (EM). Among

them, BW [3] is a common noise in ECG recording, which can be caused by respiration, electrode impedance change and motion. BW may mask important information about ECG, and even lose key diagnostic information contained in ECG. MA [4] is a muscle artifact signal generated by skeletal muscle activity, which occupies the same spectral space as the signal generated by the heart, which is also a common noise in long-range ECG. Because the frequency spectrum of MA overlaps with that of real ECG, simple filtering methods cannot achieve an effective denoising effect. EM [5] is caused by changes in electrode impedance and skin potential, and its duration is short, resulting in small changes in ECG waveform morphology. Due to a wide frequency distribution range of noises, it is difficult to obtain a good denoising effect by general denoising methods.

As for ECG denoising, one kind of common denoising method is low-pass filtering or high-pass filtering. Among them, low-pass filtering is used to filter high-frequency components in ECG signals. Many researchers set different cutoff frequencies to filter power frequency interference and other high-frequency noise in the signal and the commonly used cutoff frequencies include 30 Hz [6], 50 Hz [7], and 70

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Hz [8]. On the contrary, high pass filtering is used to filter low-frequency components such as BW in ECG signals. Common cutoff frequencies set used by researchers include 0.5 Hz [9,10], 1 Hz [11] and 5 Hz [12]. In addition to low-pass filters and high-pass filters, band-pass filters can also be used to denoise ECG signals. Afkhami et al. [13] used bandpass filtering with a filtering range of 0.1–100 Hz. In addition, two-stage median filtering was used to remove BW in the signal. Gutta et al. [14] used a fourth-order Butterworth bandpass filter with a cutoff frequency of 1–40 Hz to remove noise from the original ECG signal. And then, many researchers have tried some other classical linear filters, such as adaptive filter [15] and filter bank [16], as well as other filtering methods, including minimum mean square [17], recursive mean square [18] and Kalman filter [19–22]. Using time domain signal decomposition technology [23,24] is another common type of ECG denoising method, such as using discrete wavelet transform (DWT) [25]. In essence, DWT-based approach decomposes the signal into different frequency bands by using wavelet transform (WT) and calculates the approximation coefficient (cA) and detail coefficient (cD). By processing and reconstructing the wavelet coefficients of each layer, different levels of signal denoising can be realized. Ercelebi [26] adopted the lifting wavelet filter to obtain the wavelet coefficients of ECG signals and realized the fast denoising of MA, EM and white noise. Thomas et al. [27] used DB2 wavelet to decompose the ECG signal in five layers, calculated its d1–d5 cDs and A5 cAs, and utilized the cDs of the fourth and fifth layers to reconstruct the signal, so as to obtain the denoised signal. Shi et al. [28] used db6 wavelet to decompose the ECG signal in eight layers and utilized the cDs from the second layer to the eighth layer to reconstruct the signal for denoising the original ECG signal. However, these mentioned traditional filtering methods often only focus on one or two kinds of noise. In the acquisition process, when encountering different ECG noises, different denoising methods should be used for ECG denoising. The abovementioned limitations lead to insufficient generalization ability of the existing denoising methods under different noise conditions. Then, in the actual applications, there are many kinds of noise, so it is difficult for a single traditional filtering method to obtain a good denoising effect for all kinds of noise. Thus, this paper explores a data-driven denoising method.

The abovementioned analysis shows that the existing ECG denoising methods still have room for further improvement. With the emergence of deep learning approaches, many researchers have explored the application of deep learning to ECG signal denoising [29,30]. Recently, the denoising algorithm based on the de-noising auto-encoder (DAE) has been proven to have better performance than the traditional denoising algorithm. Considering signal denoising and signal compression simultaneously, DAE helps reduce the cost and improve the efficiency of signal processing. Pongponsoi et al. [31] proposed an adaptive ECG signal denoising method based on DWT and artificial neural network by combining the multi-resolution characteristics of wavelet decomposition and the adaptive learning ability of artificial neural network. The experimental results showed that this method could successfully remove a large range of noise and significantly improve the signal-to-noise ratio. Xiong et al. [32] utilized WT and scale adaptive threshold method to filter most of the noise and then removed other residual noise based on the improved DAE. The experimental results showed that this method has significantly improved signal-to-noise ratio (SNR) and root mean square error (RMSE). Chiang et al. [29] proposed an ECG signal denoising method based on full convolution network (FCN), which used the advantages of FCN to process ECG signals. Arsene et al. [30] designed a convolutional neural network (CNN) including six convolution layers. A pooling layer, corresponding to the convolution layer and a full connection layer for regression, were applied to ECG signal denoising. Singh et al. [33] adopted the generative countermeasure network (GAN) model based on CNN to achieve end-to-end effective denoising of noise data. Pascal et al. [49] designed encoder and decoder, called DAN, to recognize partial noise in input data. However, when using deep learning technology to process ECG signal denoising, simple

down-sampling processing in deep learning-based methods will lead to the loss of effective information when the network is compressed, which will have a negative impact on the final denoising effect.

Considering these limitations and problems, this paper proposes a signal denoising method based on CNN combining WT, called a deep wavelet convolutional neural network (DW-CNN), to further improve the denoising results. The input of this method is a whole segment of ECG signal with different noise types. Meanwhile, using the structure of the DAE for reference, the convolution layers are used to replace the simple full-connected layers to build the encoder and decoder of the signal. To overcome the problem of simply discarding the input information in the process of signal down-sampling caused by the pooling layer, DWT is used to convert the signal into high-frequency and low-frequency components for modifying the pooling layer in down-sampling the signal, so as to compress the signal length and retain all the effective information. Finally, for verifying the effectiveness of DW-CNN, the datasets under real noise conditions based on 10 different long-range ECG data are utilized for evaluating the proposed method.

Further, the innovations of this paper are listed as follows.

- 1) The proposed DW-CNN avoids QRS waveform detection, which can avoid iterative errors caused by QRS complex detection. Additionally, there is no need to perform any other normalization operation to simplify the preprocess process.
- 2) In this paper, CNN is used as an automatic feature extractor to build an encoder and decoder, which can effectively automatically extract numerous deep features and achieve end-to-end automatic noise reduction. This end-to-end method reaches better denoising performance and generalization ability.
- 3) For overcoming the problem of simply discarding the input information caused by the pooling layer, this paper utilizes DWT and IDWT to build DWT-based pooling layer and IDWT-based up-sampling layer, which can retain all the effective information and improve the denoising performance.

The remainder of this paper is described as follows. Section II presents the evaluation datasets under different noise conditions. Section III proposes the methodology based on CNN and DWT-based pooling layers. Detailed descriptions of the ablation experiments and comparison experiments and the experimental results are shown in Section IV. Meanwhile, we discuss the experimental results. Finally, Section V concludes the proposed DW-CNN.

2. Materials

As the verification standard of the denoising algorithm proposed in this paper, the Massachusetts Institute of Technology Beth Israel Hospital (MIT-BIH) arrhythmia database (MITDB) [34] recorded a total of 48 records of 47 sampled people collected by Beth Israel Hospital from 1975 to 1979. The ECG signal is collected by a sampling device with a sampling frequency of 360 Hz and a resolution of 11 bits. Since lead II data is widely used in ECG applications based on mobile or wireless somatosensory networks [35], lead II data from 10 clean ECG records in MITDB is selected as the clean dataset without noise in this paper. Additionally, the real noise data obtained from the Massachusetts Institute of Technology Beth Israel Hospital (MIT-BIH) noise stress test database (NSTDB) [36] includes BW, MA and EM. NSTDB contains a variety of different noise data, such as BW, EM, and MA. Meanwhile, the database provides the nst data processing script, which can generate the corresponding noisy ECG based on the clean ECG record according to the set parameters.

$$s_{out} = s_{clean} + a * noise + b \quad (1)$$

Where a is gain, b is offsets, s_{clean} means input clean ECG and s_{out} means output ECG.

Table 1
Details of Different Datasets.

Dataset	Record	Lead	Number
MITDB	103	MLII	1321
	105		1394
	111		1407
	116		1131
	122		1366
	205		1403
	213		1205
	219		1323
	223		1344
	230		1287
NSTDB	103	MLII	1321
	105		1394
	111		1407
	116		1131
	122		1366
	205		1403
	213		1205
	219		1323
	223		1344
	230		1287

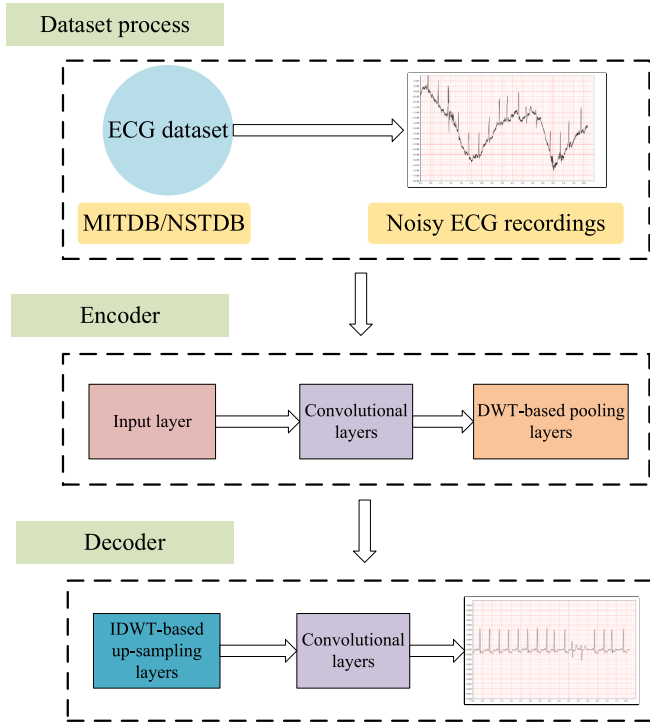


Fig. 1. Overview of DW-CNN.

To further evaluate the generalization ability of the denoising model in different noise environments [29,32], three real noise sources in NSTDB are synthesized at different signal-to-noise ratios (including 0 dB, 1.25 dB and 5 dB) to obtain ECG signal datasets with different noises via the proposed processing script in NSTDB. And then, in order to build the evaluation ECG dataset required for the experiment, the 10 clean ECG recordings collected in MITDB and the synthesized noisy ECG records are uniformly segmented to obtain the corresponding ECG signals, and the sampling length of each ECG signal is 4096 sampling points. Then, Table 1 describes the details of each ECG signal under different noise environments.

3. Methodology

3.1. Overview of DW-CNN

Fig. 1 shows the flow chart of the proposed DW-CNN in this paper. The original signal is directly fed into the network input layer and then processed by the denoising mode based on the deep neural network (DNN). It is worth noting that there is no special structure in the input layer, which represents the entrance of the deep network and is responsible for receiving the noisy ECG signal. Finally, the final denoised ECG signal is obtained by the output of DW-CNN. The detailed introduction of the proposed DW-CNN is described in the following sections.

3.2. Pre-processing

This Section shows the proposed process of preprocessing stage. Compared with the denoising model based on heartbeat [37], this paper is devoted to the denoising of the whole ECG signal. The heartbeat-based denoising model firstly requires the accurate QRS waveform position and then cuts out each heartbeat from the ECG signal. In practical applications, QRS complex detection algorithms [38] have been studied by many researchers. However, one of the main problems is that the QRS complex detection algorithm cannot ensure good robustness in the noisy environment, resulting in inaccurate reference R-peak points, which brings cumulative errors to subsequent analysis.

Therefore, this paper directly denoises the original ECG signal to avoid waveform detection. In addition, at the stage of inputting the original data, there is no need to perform any other normalization operation, which can avoid the distortion of the amplitude information of the ECG signal and retain the original characteristics of the ECG signal.

Comprehensively considering the signal acquisition time, denoising processing time, denoising effect and final data storage, the ECG signals at 4096 sampling points are taken as the research objects. When the database is used to verify the denoising effect of the model, the original data is divided into 4096 ECG samples. Notably, due to the different sampling frequencies of different acquisition devices, the acquisition time will be slightly different. Take most of the dynamic ECG signal acquisition devices used as an example, which can collect ECG signals for about 8–10 s. Fig. 2 shows an example of a clean ECG signal and a corresponding noisy ECG signal.

3.3. Convolutional neural network (CNN)

Fukushima [39] firstly proposed a CNN structure for pattern recognition. With continuous development in recent years, CNN has developed into a common DNN structure. Compared with the fully-connected network, CNN realizes the sharing of parameter weights, improves the efficiency of parameter utilization, makes the model easier to train, and further avoids the overfitting situation in the training of the model. CNN mainly includes the input layer, convolution layer, ReLU layer (activation layer), pooling layer and full connected layer.

The convolution layer is the core process of CNN, which is utilized for extracting deep features for the training dataset. Equation (2) presents a convolution process with convolution kernel size (k) and the stride (k).

$$o_i = \sigma \left(\sum_{j=i}^{i+k} w_j x_j + b \right) \quad (2)$$

Where x_j is input, w_j is weight, b means bias, $\sigma(\cdot)$ means Relu function and o_i means output.

By combining and stacking the above layers, a complete CNN can be constructed. In addition, according to the actual task requirements, the corresponding output loss function is constructed, and the loss function is optimized according to the corresponding optimizer, so as to adjust

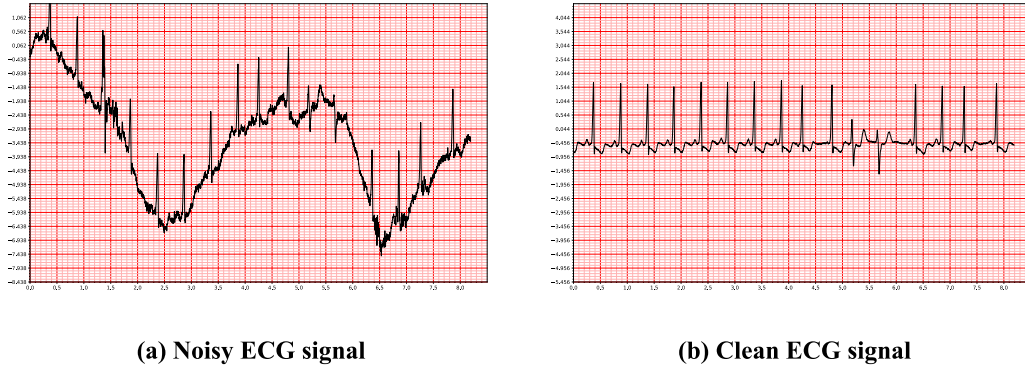


Fig. 2. An example of clean ecg signal and corresponding noisy ecg signal.

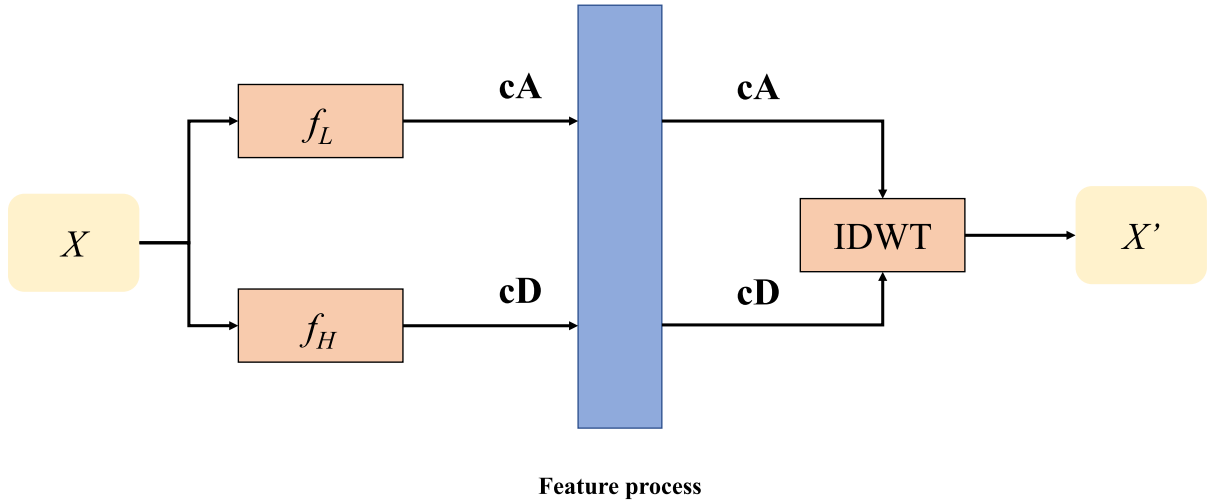


Fig. 3. The diagram of signal decomposition and reconstruction.

various parameters in the network model. In recent years, researchers use CNN in different fields and achieve good results, including clinical detection [40–43], fault diagnosis [44–46], geological formation recognition [47], and so on. Thus, this paper selected CNN to build as the basic structure of the proposed method for acquiring better performance.

3.4. Dwt-based pooling layer and IDWT-based up-sampling layer

Generally, CNN adopts the maximum pooling strategy for sampling to perform the pooling layer operation, which is used to reduce the spatial size of data and thus reduce the number of parameters of the overall network. However, in the maximum pooling operation, the pooling layer directly selects the maximum value in the pooling window to characterize the data characteristics covered by this pooling window. However, the abovementioned pooling operation inevitably brings about certain information loss, which may lead to the loss of some information beneficial to the final prediction result in the training process, and then affect the final model performance. Therefore, in this paper, DWT is introduced to the deep learning-based model structure. Since DWT is reversible, this pooling scheme can ensure that all feature information is retained. In addition, DWT can obtain the cD and cA of the signal, which helps to save various detailed information about the signal.

$$X_1 = f_L \otimes X \downarrow_2 \quad (3)$$

$$X_2 = f_H \otimes X \downarrow_2 \quad (4)$$

Where f_L is low-pass filter, f_H is high-pass filter, X means input, and X_1 and X_2 mean output after DWT. Where X_1 represents the cA and X_2 represents the cD.

Taking Haar wavelet as an example, the relationship between DWT and pooling operation is explained here. In the one-dimensional Haar wavelet, the low-pass filter f_L and the high-pass filter f_H are defined as follows.

$$f_L = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right), f_H = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right) \quad (5)$$

It can be seen from Equation (3) that it is a pooling operation. When only low-frequency subbands are considered, DWT and inverse discrete wavelet transform (IDWT) play the roles of pooling layer and up-sampling layer in the network respectively. Further, when all subbands are considered, the DWT-based pooling operation can take all information of the signal into account, which is helpful for signal reconstruction. Compared with the maximum pooling operation, since DWT has biorthogonal characteristics, the original signal $\times$ can be reconstructed more accurately by IDWT. Therefore, DWT-based pooling layer is used to replace the conventional maximum pooling layer.

Fig. 3 shows the diagram of signal decomposition and reconstruction using DWT. In the decomposition stage, two predefined filters are deployed for the input data at each depth, and then predefined wavelet functions are used as pooling operators. In the reconstruction stage, each sub-band image is upsampled, that is, the data is reconstructed and restored using IDWT to obtain the current level of reconstruction results X' .

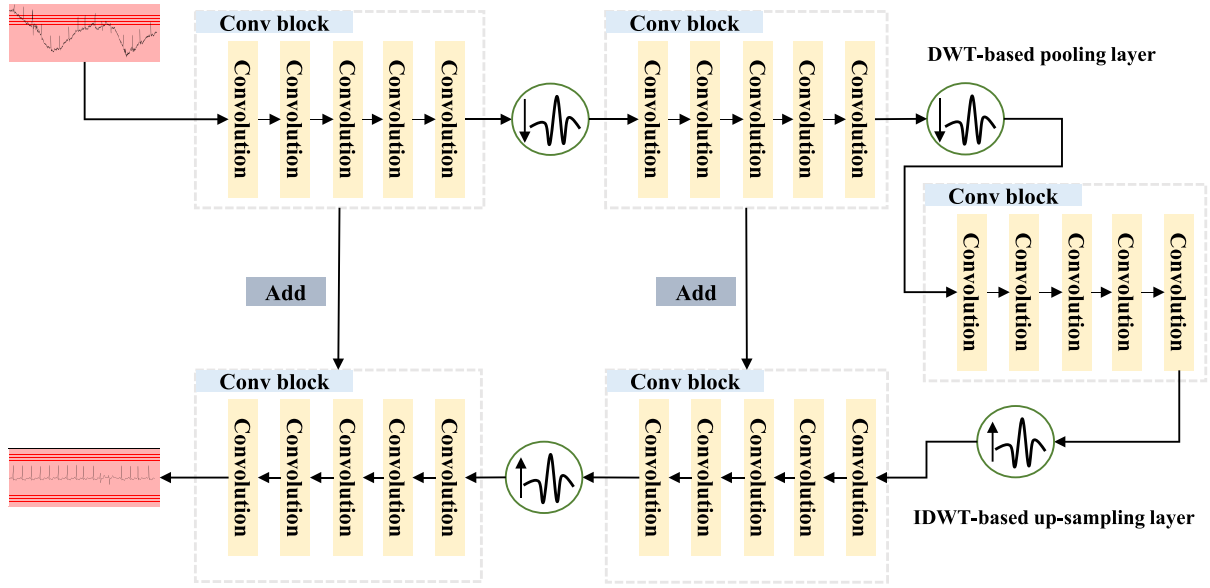


Fig. 4. The architecture of DW-CNN.

Table 2
Details of DW-CNN.

Layers	Type	Kernel size	Stride	Output size
0–1	Input	–	–	1*4096
1–2	Convolution	3	1	64*4096
2–3	Convolution	3	1	64*4096
3–4	Convolution	3	1	64*4096
4–5	Convolution	3	1	64*4096
5–6	Convolution	3	1	64*4096
6–7	DWT-based pooling layer	–	–	128*2048
7–8	Convolution	3	1	64*2048
8–9	Convolution	3	1	64*2048
9–10	Convolution	3	1	64*2048
10–11	Convolution	3	1	64*2048
11–12	Convolution	3	1	64*2048
12–13	DWT-based pooling layer	–	–	128*1024
13–14	Convolution	3	1	64*1024
14–15	Convolution	3	1	64*1024
15–16	Convolution	3	1	64*1024
16–17	Convolution	3	1	64*1024
17–18	Convolution	3	1	64*1024
18–19	Convolution	3	1	128*1024
19–20	IDWT-based up-sampling layer	–	–	64*2048
20–21	Add layer	–	–	64*2048
21–22	Convolution	3	1	64*2048
22–23	Convolution	3	1	64*2048
23–24	Convolution	3	1	64*2048
24–25	Convolution	3	1	64*2048
25–26	Convolution	3	1	128*2048
26–27	IDWT-based up-sampling layer	–	–	64*4096
27–28	Add layer	–	–	64*4096
28–29	Convolution	3	1	64*4096
29–30	Convolution	3	1	64*4096
30–31	Convolution	3	1	64*4096
30–31	Convolution	3	1	1*4096

3.5. The architecture of DW-CNN

Fig. 4 depicts the structure of DW-CNN. The network structure is composed of two parts, encoder and decoder, which are composed of convolution layer, DWT-based pooling layer or IDWT-based up-sampling layer.

Firstly, convolution block is composed of convolution layers, each convolution block is composed of 5 or 6 convolution layers. The convolution layer mainly performs various nonlinear processing on the

training data and then excavates the deep-seated hidden features contained in the noisy ECG signal data, which is helpful for the encoding, reconstruction and decoding of waveform signals. It is worth noting that since the ECG signal is a one-dimensional discrete signal, a one-dimensional convolution layer is used here. On the one hand, the DWT-based pooling layer can reduce the spatial size of the signal and the amount of model parameter calculation, which is similar to the common pooling layer. On the other hand, to overcome the situation that the common pooling layer may lose part of the information, resulting in incomplete information and inaccurate construction during up-sampling, DWT-based pooling layer converts the input data into cDs and cAs through DWT, thereby saving all the information of the input data and ensuring the comprehensiveness of the information. The IDWT-based up-sampling layer is to up-sample the data to reconstruct the denoised signal. Compared with the conventional-based up-sampling layer, the IDWT-based up-sampling layer can fully utilize all the information in the data and ensure the accuracy of the reconstructed data.

Table 2 shows the details of DW-CNN. In the input layer, the input length of the ECG signal is 1×4096 . In the encoder stage, the input one-dimensional ECG signal is digitally encoded using convolution blocks consisting of convolution layers with 64/128 filters and DWT-based pooling layers, with an output size of 128×1024 . Subsequently, in the decoder stage, the output results of the encoder are processed non-linearly using the convolution blocks composed of convolution layers and IDWT-based up-sampling layers implementation. Finally, the elementwise operation is used to merge the up-sampled feature map with the feature map after the input layer to complete the decoding of the encoder output features, whose output size is the same as that of the input layer, 1×4096 .

It is noted that in the convolution block, we introduce the padding technique, that is, one feature point is padded before and after the feature, for ensuring that the last dimension feature is unchanged after the processing of the convolution layer.

3.6. Model training

This section introduces the loss functions and training parameters required in the network training process. For optimizing the network structure parameters, the mean square error (MSE) function is used as the loss function in the training process to represent the effect of the training process. Equation (6) represents the definition of the MSE function.

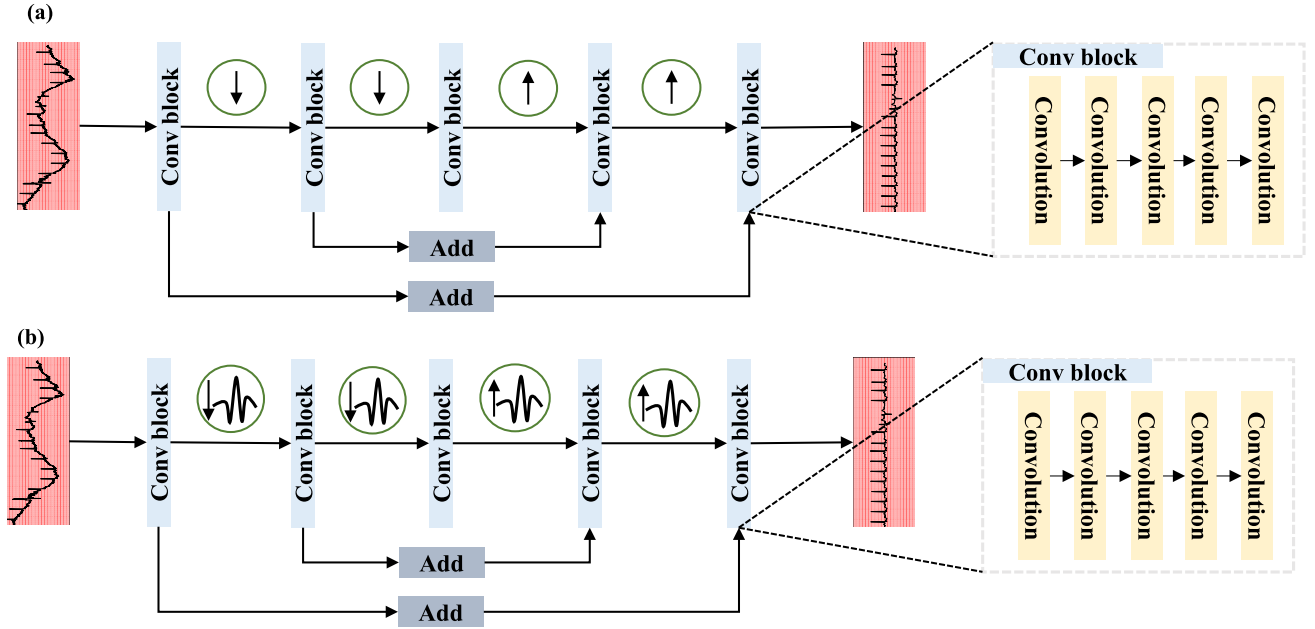


Fig. 5. The network structures used in ablation experiments.

Table 3
Ablation experiments with 0 dB noise intensities under BW.

Record		103	105	111	116	122	205	213	219	223	230	Average
modelNet1	RMSE	0.162	0.327	0.257	0.7	0.345	0.217	0.265	0.422	0.214	0.303	0.321
db2	RMSE	0.158	0.331	0.272	0.672	0.308	0.211	0.257	0.406	0.205	0.326	0.315
db3	RMSE	0.159	0.331	0.275	0.675	0.316	0.193	0.253	0.407	0.202	0.327	0.314
db4	RMSE	0.164	0.335	0.258	0.669	0.327	0.2	0.278	0.421	0.213	0.339	0.32
db5	RMSE	0.159	0.339	0.271	0.665	0.305	0.196	0.254	0.41	0.202	0.292	0.309
db6	RMSE	0.149	0.334	0.262	0.655	0.297	0.194	0.244	0.406	0.206	0.277	0.302
coif2	RMSE	0.192	0.326	0.272	0.715	0.404	0.162	0.27	0.432	0.22	0.296	0.329
coif3	RMSE	0.153	0.336	0.277	0.661	0.311	0.195	0.243	0.406	0.202	0.265	0.305
coif4	RMSE	0.156	0.354	0.283	0.668	0.306	0.206	0.256	0.412	0.207	0.279	0.313
coif5	RMSE	0.151	0.335	0.254	0.664	0.287	0.172	0.235	0.406	0.198	0.28	0.298
Haar	RMSE	0.165	0.324	0.254	0.646	0.291	0.16	0.224	0.398	0.195	0.28	0.294
modelNet1	SNR	8.316	6.152	1.946	5.801	9.299	5.174	8.724	6.351	10.551	1.453	6.377
db2	SNR	9.846	5.937	1.704	7.234	10.802	5.363	9.159	6.834	10.87	0.913	6.866
db3	SNR	9.136	5.751	1.474	6.543	10.323	6.23	9.469	6.865	11.389	1.051	6.823
db4	SNR	9.171	5.938	1.928	6.564	9.808	6.344	8.375	6.368	10.834	0.465	6.58
db5	SNR	9.329	5.537	1.71	7.12	10.66	6.14	9.668	6.758	10.835	1.882	6.964
db6	SNR	9.809	5.876	1.83	7.23	10.892	6.22	9.629	6.883	10.888	2.257	7.151
coif2	SNR	6.129	5.843	1.229	4.822	7.394	7.956	9.349	6.032	10.286	1.923	6.096
coif3	SNR	9.935	5.508	1.586	7.219	10.367	6.323	10.06	6.858	11.23	2.75	7.184
coif4	SNR	9.206	4.83	1.225	6.682	10.659	5.865	9.611	6.764	10.793	2.225	6.786
coif5	SNR	9.5	6.039	2.145	7.521	11.861	7.789	10.305	6.99	11.271	2.45	7.587
Haar	SNR	10.186	5.868	2.351	7.691	11.345	8.36	10.657	7.122	11.306	2.24	7.713

$$Loss = \frac{1}{N} \sum_{i=1}^N (p(x_i) - q(x_i))^2 \quad (6)$$

Where $p(x_i)$ is the clean ECG result, $q(x_i)$ is the denoised ECG result, and N is the sampling length of the ECG signal.

In the training process, Adam [48] optimizer is used to optimize the target loss function. Equation (7) shows an overview of how the optimizer updates parameters.

$$\theta_i = \theta_{i-1} - \alpha \frac{\partial Loss}{\partial \theta_{i-1}} \quad (7)$$

Where θ_i means parameter weight at i epoch. Meanwhile, considering the convergence speed of the model and the memory of the workstation, 512 is selected as the size of each batch during training.

4. Results and discussion

As described above, the noise clean ECG dataset is constructed by using MITDB and NSTDB datasets, and 80%, 10% and 10% of the data in each noise record are randomly selected for synthesis and used as the training set, validation set and test set. The training set is used to train the model, the validation set is used to select the best model, and the test set is only used to evaluate the model performance to ensure that the data information is not leaked.

To fully evaluate the reliability of DW-CNN, we design experiments to verify the proposed denoising model from the following two aspects.

- 1) Ablation experiments. The purpose of ablation experiments is to verify that the structure of DW-CNN is reasonable and reliable. In the structure of DW-CNN, there are mainly two factors that will affect the final structure: compared with the conventional maximum pooling

Table 4

Ablation experiments with 1.25 dB noise intensities under BW.

Record		103	105	111	116	122	205	213	219	223	230	Average
modelNet1	RMSE	0.21	0.352	0.294	0.754	0.257	0.232	0.311	0.398	0.198	0.287	0.329
db2	RMSE	0.178	0.316	0.245	0.783	0.306	0.162	0.38	0.401	0.244	0.253	0.327
db3	RMSE	0.151	0.319	0.225	0.786	0.364	0.182	0.369	0.432	0.262	0.299	0.339
db4	RMSE	0.232	0.331	0.277	0.797	0.336	0.167	0.327	0.397	0.204	0.274	0.334
db5	RMSE	0.155	0.317	0.229	0.742	0.291	0.193	0.352	0.398	0.216	0.297	0.319
db6	RMSE	0.136	0.313	0.203	0.741	0.298	0.137	0.32	0.393	0.222	0.24	0.3
coif2	RMSE	0.167	0.319	0.243	0.766	0.333	0.179	0.336	0.427	0.235	0.298	0.33
coif3	RMSE	0.156	0.315	0.243	0.775	0.31	0.173	0.307	0.401	0.215	0.261	0.316
coif4	RMSE	0.142	0.311	0.215	0.76	0.35	0.16	0.304	0.418	0.26	0.26	0.318
coif5	RMSE	0.158	0.323	0.241	0.721	0.251	0.177	0.301	0.384	0.193	0.267	0.302
Haar	RMSE	0.126	0.299	0.21	0.753	0.315	0.131	0.291	0.401	0.225	0.24	0.299
modelNet1	SNR	6.328	4.432	0.539	5.889	11.883	4.424	8.138	6.606	11.081	3.818	6.314
db2	SNR	8.391	6.892	2.735	5.477	10.455	7.983	6.325	6.628	9.908	5.305	7.01
db3	SNR	9.641	6.982	3.478	5.616	8.572	7.155	6.071	5.794	8.749	3.614	6.567
db4	SNR	5.158	5.391	0.625	4.876	9.148	8.054	7.478	6.719	11.281	4.365	6.31
db5	SNR	9.066	6.88	3.334	6.553	10.523	6.48	6.768	6.669	10.675	3.459	7.041
db6	SNR	10.579	7.417	4.652	6.841	10.642	9.313	7.584	7.097	10.252	6.008	8.038
coif2	SNR	8.813	6.964	2.709	5.824	9.198	7.197	6.931	5.94	9.77	3.693	6.704
coif3	SNR	8.812	6.771	2.686	5.655	10.005	7.615	7.983	6.69	10.497	5.09	7.18
coif4	SNR	9.763	7.6	3.917	6.024	8.664	8.145	7.928	6.257	8.738	5.195	7.223
coif5	SNR	9.086	6.694	2.849	7.292	12.353	7.241	8.228	7.186	11.514	4.587	7.703
Haar	SNR	11.156	7.948	4.301	6.716	10.252	9.543	8.545	6.777	10.077	6.093	8.141

Table 5

Ablation experiments with 5 dB noise intensities under BW.

Record		103	105	111	116	122	205	213	219	223	230	Average
modelNet1	RMSE	0.193	0.3	0.202	0.506	0.304	0.205	0.311	0.359	0.228	0.297	0.291
db2	RMSE	0.22	0.32	0.236	0.441	0.259	0.222	0.314	0.336	0.2	0.325	0.287
db3	RMSE	0.203	0.305	0.224	0.493	0.275	0.226	0.298	0.339	0.21	0.299	0.287
db4	RMSE	0.218	0.324	0.23	0.451	0.261	0.23	0.314	0.343	0.204	0.32	0.29
db5	RMSE	0.211	0.306	0.219	0.483	0.26	0.191	0.292	0.329	0.201	0.285	0.278
db6	RMSE	0.194	0.297	0.2	0.503	0.277	0.156	0.27	0.333	0.204	0.244	0.268
coif2	RMSE	0.219	0.318	0.224	0.502	0.283	0.217	0.331	0.361	0.217	0.323	0.3
coif3	RMSE	0.199	0.309	0.23	0.542	0.378	0.16	0.316	0.395	0.259	0.285	0.307
coif4	RMSE	0.18	0.298	0.203	0.523	0.327	0.143	0.278	0.342	0.225	0.226	0.274
coif5	RMSE	0.19	0.305	0.209	0.515	0.288	0.146	0.275	0.336	0.193	0.247	0.27
Haar	RMSE	0.221	0.316	0.225	0.428	0.238	0.203	0.29	0.327	0.179	0.319	0.275
modelNet1	SNR	7.979	6.281	4.325	7.01	9.958	6.217	7.974	7.564	10.657	2.603	7.057
db2	SNR	7.141	5.054	3.071	8.247	11.213	5.941	8.278	8.344	11.923	1.943	7.115
db3	SNR	7.755	5.611	3.651	7.165	10.745	5.25	8.391	8.255	11.54	2.572	7.094
db4	SNR	7.263	4.837	3.262	7.992	11.109	5.481	8.192	8.239	11.897	1.927	7.02
db5	SNR	7.61	5.508	3.728	7.422	11.167	8.373	8.938	8.54	12.164	3.211	7.666
db6	SNR	7.687	5.763	4.041	6.793	10.516	9.733	9.281	8.533	11.766	4.402	7.852
coif2	SNR	7.061	5.245	3.377	6.998	10.404	5.952	7.529	7.55	11.436	1.757	6.731
coif3	SNR	7.16	5.703	2.888	6.054	7.756	9.949	7.833	6.422	9.472	3.086	6.632
coif4	SNR	7.811	6.13	3.542	6.256	9.076	9.695	8.979	8.302	11.095	5.232	7.612
coif5	SNR	7.677	5.324	3.568	6.469	10.159	10.505	9.314	8.414	12.673	4.455	7.856
Haar	SNR	7.282	4.907	3.467	8.783	12.136	7.593	9.227	8.597	12.769	2.132	7.689

layer, the DWT-based pooling layer can achieve performance advantages or not; For the DWT-based pooling layer, the selection of wavelet type will affect the final performance or not. Fig. 5 shows various network structures used in ablation experiments. Fig. 5 (a) shows that the maximum pooling operation is used to replace the DWT-based pooling layer and the conventional-based up-sampling layer to replace the IDWT-based up-sampling layer (modelNet1), to realize the encoding and decoding process of network data. In Fig. 5 (b), Daubechies (db, db2-db6) wavelet, coiflets (coif, coif2-5) wavelet and Haar wavelet are selected as wavelet types for verification to research the impact of wavelet types on the final experimental results. It is worth noting that in the ablation experiments, in addition to the above changes, the structure of other encoders and decoders, the hyperparametric settings are consistent for ensuring the fairness of the experiment.

2) Comparative experiments. The existing ECG denoising methods need to perform maximum and minimum normalization processing on the training data. However, this normalization processing is not conducive to subsequent analysis and is prone to data information leakage,

which may lead to inconsistency between the experimental results and the actual application results. Thus, to further compare with the existing methods, according to the model structure described in the article [29,49], the FCN and the DNN-based DAN are reproduced and verified on the test set. Similar to the ablation experiment settings, except for the change of network structure, the super-parameter settings during the training process are consistent, to ensure the fairness and rationality of the comparison experiment.

This paper uses the internationally commonly used evaluation metrics, RMSE and SNR, to evaluate the model denoising performance. Equation (8) and Equation (9) define RMSE and SNR respectively.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (x(i) - x_m(i))^2} \quad (8)$$

$$SNR = 10 \log \frac{\sum_{i=1}^N (x(i))^2}{\sum_{i=1}^N (x(i) - x_m(i))^2} \quad (9)$$

Table 6

Ablation experiments with 0 dB noise intensities under EM.

Record		103	105	111	116	122	205	213	219	223	230	Average
modelNet1	RMSE	0.285	0.252	0.212	0.49	0.406	0.199	0.394	0.472	0.315	0.337	0.336
db2	RMSE	0.282	0.254	0.218	0.397	0.362	0.189	0.385	0.458	0.311	0.336	0.319
db3	RMSE	0.288	0.258	0.226	0.419	0.36	0.194	0.385	0.445	0.303	0.329	0.321
db4	RMSE	0.277	0.253	0.209	0.454	0.381	0.187	0.398	0.465	0.311	0.353	0.329
db5	RMSE	0.288	0.255	0.213	0.399	0.353	0.171	0.391	0.449	0.298	0.347	0.316
db6	RMSE	0.278	0.247	0.216	0.42	0.354	0.158	0.386	0.451	0.297	0.339	0.315
coif2	RMSE	0.32	0.268	0.243	0.558	0.437	0.158	0.41	0.475	0.314	0.346	0.353
coif3	RMSE	0.301	0.261	0.223	0.399	0.332	0.163	0.382	0.441	0.293	0.352	0.315
coif4	RMSE	0.303	0.27	0.225	0.461	0.339	0.162	0.404	0.446	0.308	0.376	0.329
coif5	RMSE	0.283	0.257	0.207	0.404	0.358	0.142	0.39	0.449	0.297	0.326	0.311
Haar	RMSE	0.293	0.259	0.217	0.388	0.347	0.157	0.378	0.442	0.293	0.34	0.311
modelNet1	SNR	5.922	5.145	3.698	7.488	8.35	5.395	6.239	5.656	7.72	1.756	5.737
db2	SNR	6.608	5.264	3.778	10.409	10.054	6.12	6.503	5.991	7.463	1.715	6.391
db3	SNR	6.28	5.229	3.293	9.259	9.688	5.779	6.739	6.289	7.876	2.105	6.254
db4	SNR	6.75	5.286	3.777	8.502	8.765	6.023	6.132	5.686	7.799	1.336	6.006
db5	SNR	6.431	5.414	3.942	9.872	9.946	7.001	6.834	6.241	8.191	1.863	6.574
db6	SNR	6.713	5.611	3.709	9.275	9.964	7.802	6.835	6.336	8.512	2.173	6.693
coif2	SNR	3.506	4.402	1.801	5.917	6.752	8.242	6.094	5.375	7.74	2.346	5.217
coif3	SNR	5.934	5.087	4.083	9.889	10.444	8.229	7.171	6.404	8.632	1.803	6.768
coif4	SNR	5.429	4.744	3.573	8.222	10.156	8.058	6.616	6.224	8.247	1.273	6.254
coif5	SNR	6.362	5.456	4.348	10.125	10.658	9.461	7.05	6.36	8.622	2.574	7.102
Haar	SNR	6.745	5.314	4.182	10.511	10.507	8.895	7.294	6.442	8.541	2.176	7.061

Table 7

Ablation experiments with 1.25 dB noise intensities under EM.

Record		103	105	111	116	122	205	213	219	223	230	Average
modelNet1	RMSE	0.275	0.307	0.238	0.541	0.339	0.188	0.365	0.457	0.265	0.337	0.331
db2	RMSE	0.258	0.282	0.2	0.568	0.376	0.138	0.345	0.463	0.289	0.278	0.32
db3	RMSE	0.261	0.29	0.184	0.578	0.423	0.162	0.381	0.504	0.306	0.28	0.337
db4	RMSE	0.298	0.319	0.241	0.59	0.412	0.151	0.376	0.457	0.262	0.324	0.343
db5	RMSE	0.261	0.287	0.194	0.512	0.347	0.195	0.372	0.464	0.267	0.313	0.321
db6	RMSE	0.262	0.291	0.188	0.511	0.343	0.133	0.343	0.457	0.272	0.264	0.306
coif2	RMSE	0.267	0.287	0.199	0.551	0.388	0.149	0.363	0.48	0.291	0.304	0.328
coif3	RMSE	0.272	0.29	0.205	0.545	0.359	0.148	0.352	0.465	0.277	0.297	0.321
coif4	RMSE	0.274	0.292	0.193	0.534	0.396	0.148	0.351	0.47	0.301	0.284	0.324
coif5	RMSE	0.274	0.295	0.205	0.493	0.315	0.145	0.364	0.454	0.258	0.307	0.311
Haar	RMSE	0.262	0.279	0.189	0.51	0.362	0.133	0.333	0.46	0.266	0.272	0.307
modelNet1	SNR	6.307	5.788	2.723	7.24	9.704	6.242	7.357	6.325	9.292	1.527	6.251
db2	SNR	7.911	7.543	4.364	7.322	8.789	9.242	7.642	6.158	8.557	3.327	7.086
db3	SNR	6.996	6.757	4.75	6.766	7.724	7.228	6.371	5.092	8.039	3.225	6.295
db4	SNR	4.36	4.965	1.984	6.068	7.28	8.835	7.052	6.406	9.718	1.95	5.862
db5	SNR	7.225	7.141	4.496	8.254	9.673	5.605	6.824	6.273	9.349	2.241	6.708
db6	SNR	7.547	7.261	5.013	8.834	10.469	9.394	7.773	6.585	9.559	4.164	7.66
coif2	SNR	7.387	7.354	4.257	7.231	8.283	8.386	7.164	5.621	8.74	2.448	6.687
coif3	SNR	6.732	7.188	4.069	7.265	9.248	8.64	7.74	6.223	8.872	2.881	6.886
coif4	SNR	6.665	6.536	4.589	7.645	8.019	8.304	7.288	5.867	8.066	3.276	6.626
coif5	SNR	7.444	7.239	4.227	9.183	11.662	8.738	7.618	6.721	9.597	2.588	7.502
Haar	SNR	7.266	7.518	5.006	8.575	9.706	9.524	8.048	6.393	9.528	3.857	7.542

Where $x(i)$ and $x_m(i)$ represent the original ECG signal and the denoised ECG signal respectively.

4.1. Ablation experiments

In this experiment, the effectiveness of DWT-based pooling layer and the impact of wavelet type on the final result are validated in three noise environments of BW, EM and MA with different noise intensities. [Tables 3-11](#) respectively show the ablation experimental results under different noise intensity environments under BW, EM and MA noises.

To evaluate the rationality of ECG denoising model structure, detailed ablation experiments are necessary. In the ablation experiments, three noise intensities of 0 dB, 1.25 dB and 5 dB are added respectively in BW, EM and MA noise environments to comprehensively compare the denoising models based on maximum pooling operation and various denoising models using different wavelet types.

From [Tables 3-11](#), we can obtain conclusions about the structure of DW-CNN from the following three aspects.

For the overall structure, both the denoising model based on maximum pooling operation and the denoising model based on DWT can achieve a good denoising effect on noisy ECG signals. If other structures cannot extract enough effective ECG features or reconstruct the features effectively, they will not be able to effectively complete the task of ECG signal noise reduction in the end. On this basis, the results of ablation experiments strongly illustrate that the other structures of DW-CNN in this paper are reasonable.

From the operation of pooling layer, it can be seen from the results of ablation experiments that compared with the maximum pooling-based pooling layer, the DWT-based pooling layer can effectively improve the overall model denoising performance. Generally, these two evaluation metrics are interrelated. When the RMSE is smaller and the SNR is larger, the denoising model achieves better denoising results in this noise environment. In most cases, DWT-based pooling layer can achieve better denoising performance for ECG signals than the pooling layer based on maximum pooling operation. The noise type, noise intensity and waveform changes of ECG records also affect the final denoising and improvement results to a certain extent. Generally, the ablation

Table 8

Ablation experiments with 5 dB noise intensities under EM.

Record		103	105	111	116	122	205	213	219	223	230	Average
modelNet1	RMSE	0.183	0.275	0.193	0.752	0.351	0.165	0.364	0.392	0.276	0.307	0.326
db2	RMSE	0.199	0.28	0.214	0.714	0.306	0.167	0.362	0.363	0.251	0.314	0.317
db3	RMSE	0.191	0.273	0.208	0.711	0.319	0.173	0.358	0.367	0.252	0.308	0.316
db4	RMSE	0.213	0.292	0.222	0.701	0.296	0.186	0.37	0.359	0.25	0.323	0.321
db5	RMSE	0.199	0.275	0.209	0.701	0.318	0.143	0.352	0.365	0.247	0.289	0.31
db6	RMSE	0.196	0.278	0.21	0.73	0.335	0.142	0.351	0.367	0.246	0.279	0.313
coif2	RMSE	0.203	0.289	0.216	0.706	0.331	0.172	0.378	0.383	0.258	0.328	0.326
coif3	RMSE	0.192	0.284	0.207	0.831	0.467	0.134	0.372	0.437	0.307	0.287	0.352
coif4	RMSE	0.198	0.286	0.209	0.747	0.381	0.134	0.356	0.373	0.265	0.271	0.322
coif5	RMSE	0.213	0.289	0.218	0.728	0.348	0.141	0.361	0.368	0.243	0.292	0.32
Haar	RMSE	0.196	0.277	0.217	0.686	0.271	0.155	0.346	0.354	0.229	0.313	0.304
modelNet1	SNR	9.124	7.383	4.347	6.449	9.143	7.125	6.731	7.121	9.193	2.157	6.877
db2	SNR	9.797	7.349	3.704	7.761	10.429	7.609	6.959	7.967	10.081	2.207	7.386
db3	SNR	9.978	7.874	4.078	7.262	9.983	6.82	6.943	7.835	9.864	2.242	7.288
db4	SNR	9.619	6.968	3.485	7.889	10.602	6.802	6.727	8.109	10.104	1.938	7.224
db5	SNR	10.231	8.041	4.039	7.522	9.975	9.606	7.371	8.06	10.326	3.291	7.846
db6	SNR	8.199	6.897	3.702	7.202	9.32	9.741	7.345	8.114	10.35	3.564	7.443
coif2	SNR	8.91	6.845	3.551	7.17	9.349	7.293	6.45	7.201	9.728	1.564	6.806
coif3	SNR	7.534	6.594	3.352	5.11	6.128	10.1	6.312	5.721	7.819	2.983	6.165
coif4	SNR	7.425	6.522	3.252	6.785	8.108	9.597	7.086	7.935	9.557	3.766	7.003
coif5	SNR	7.457	6.069	3.219	7.191	8.913	10.168	7.027	8.072	10.446	3.493	7.205
Haar	SNR	10.127	7.607	3.778	8.609	12.158	9.141	7.588	8.473	10.95	2.184	8.061

Table 9

Ablation experiments with 0 dB noise intensities under MA.

Record		103	105	111	116	122	205	213	219	223	230	Average
modelNet1	RMSE	0.179	0.297	0.208	0.615	0.384	0.213	0.312	0.384	0.267	0.313	0.317
db2	RMSE	0.192	0.297	0.218	0.592	0.336	0.218	0.32	0.373	0.259	0.346	0.315
db3	RMSE	0.198	0.293	0.221	0.589	0.34	0.204	0.336	0.373	0.251	0.333	0.314
db4	RMSE	0.214	0.31	0.228	0.585	0.342	0.205	0.361	0.381	0.258	0.359	0.324
db5	RMSE	0.199	0.296	0.218	0.578	0.327	0.191	0.323	0.369	0.241	0.322	0.306
db6	RMSE	0.189	0.291	0.214	0.575	0.332	0.174	0.324	0.365	0.242	0.305	0.301
coif2	RMSE	0.224	0.321	0.24	0.641	0.405	0.163	0.307	0.392	0.258	0.318	0.327
coif3	RMSE	0.202	0.293	0.22	0.574	0.32	0.172	0.323	0.363	0.237	0.299	0.3
coif4	RMSE	0.229	0.317	0.23	0.576	0.331	0.173	0.357	0.372	0.26	0.314	0.316
coif5	RMSE	0.198	0.302	0.213	0.565	0.329	0.147	0.321	0.368	0.248	0.3	0.299
Haar	RMSE	0.199	0.288	0.223	0.562	0.308	0.165	0.302	0.361	0.23	0.283	0.292
modelNet1	SNR	7.942	6.245	3.584	5.916	8.783	5.007	7.887	7.272	9.002	2.085	6.372
db2	SNR	8.513	6.643	3.715	7.241	10.623	4.918	7.803	7.534	8.909	1.319	6.722
db3	SNR	7.809	6.77	3.418	6.113	10.157	5.421	7.202	7.633	9.354	1.825	6.57
db4	SNR	7.562	6.131	3.447	6.676	9.701	5.315	6.72	7.185	9.059	1.044	6.284
db5	SNR	8.026	7.062	3.863	7.422	10.614	5.945	7.816	7.714	9.69	2.119	7.027
db6	SNR	8.191	7.098	3.836	7.326	10.674	6.918	7.511	7.935	9.746	2.579	7.181
coif2	SNR	5.318	4.759	1.794	4.72	7.258	7.595	8.106	6.817	8.99	2.37	5.773
coif3	SNR	8.26	6.688	4.037	7.466	10.932	7.497	7.772	7.996	9.905	2.702	7.326
coif4	SNR	6.712	5.815	3.533	7.243	10.333	7.482	7.094	7.885	9.183	2.438	6.772
coif5	SNR	7.914	7.048	4.154	7.637	11.487	9.233	8.178	7.944	8.832	2.907	7.633
Haar	SNR	8.54	7.088	4.11	7.864	11.471	8.231	8.594	7.909	10.247	3.324	7.738

experimental results prove that the DWT-based pooling layer is meaningful and can further improve the denoising ability of ECG signals based on the CNN model structure.

In terms of wavelet types, compared with other wavelets, Haar wavelet achieves better denoising effect while maintaining the lowest computational complexity. Fig. 6 shows the time cost of different wavelet types. As seen in the Fig. 6, Haar wavelet achieves less time consumption with 0.09 s per sample, which indicates that Haar wavelet has better real-time performance in the application of wearable devices. Generally, the ablation experimental results prove that in the DWT-based pooling layer, considering the results under various noise conditions comprehensively, using Haar wavelet is the better choice to further improve the denoising ability of ECG signals.

To sum up, the results of ablation experiments under different noise environments and noise intensities show that the proposed DW-CNN is reasonable and effective, and Haar wavelet can be selected as the wavelet type in DWT-based pooling layer.

4.2. Model performance of DW-CNN (Haar wavelet)

Based on the above ablation experimental conclusions, the effectiveness of the DW-CNN is further proved by comparing the existing advanced denoising methods and DW-CNN. Before the comparative experiment, since the existing methods need to perform maximum and minimum normalization processing on the data, this processing is not conducive to subsequent analysis and is prone to data information leakage. Thus, by reproducing DNN-based DAN and FCN, the input data and other super-parameters are consistent, to ensure the fairness and rationality of the comparison experiment. Tables 12–20 respectively show the ablation experimental results under different noise intensity environments under BW, EM and MA noises.

For the evaluation of the superiority of ECG denoising models, detailed comparative experiments are necessary. In this comparative experiment, the noise intensities of 0 dB, 1.25dB and 5 dB are added respectively in three noise environments of BW, EM and Ma to comprehensively compare DNN-based DAN, FCN, and DW-CNN (Haar wavelet). From the comparison results in Tables 12–20, compared with

Table 10

Ablation experiments with 1.25 dB noise intensities under MA.

Record		103	105	111	116	122	205	213	219	223	230	Average
modelNet1	RMSE	0.234	0.31	0.237	0.551	0.347	0.192	0.383	0.437	0.266	0.313	0.327
db2	RMSE	0.209	0.273	0.196	0.669	0.422	0.133	0.446	0.471	0.311	0.273	0.34
db3	RMSE	0.188	0.307	0.199	0.606	0.408	0.168	0.411	0.464	0.3	0.308	0.336
db4	RMSE	0.23	0.334	0.246	0.6	0.395	0.155	0.389	0.433	0.264	0.289	0.334
db5	RMSE	0.225	0.292	0.206	0.524	0.371	0.204	0.421	0.444	0.282	0.346	0.332
db6	RMSE	0.181	0.288	0.184	0.551	0.364	0.13	0.388	0.428	0.275	0.255	0.304
coif2	RMSE	0.205	0.301	0.205	0.58	0.373	0.158	0.394	0.449	0.281	0.322	0.327
coif3	RMSE	0.203	0.293	0.205	0.564	0.364	0.155	0.377	0.434	0.273	0.281	0.315
coif4	RMSE	0.183	0.29	0.187	0.595	0.393	0.146	0.371	0.455	0.294	0.265	0.318
coif5	RMSE	0.212	0.304	0.212	0.503	0.331	0.145	0.384	0.426	0.272	0.298	0.309
Haar	RMSE	0.166	0.276	0.178	0.562	0.378	0.127	0.354	0.433	0.27	0.251	0.3
modelNet1	SNR	6.508	5.174	2.819	7.252	9.903	6.104	6.091	6.386	9.333	2.812	6.238
db2	SNR	6.769	7.559	4.458	6.202	7.698	9.427	5.029	5.478	7.067	3.828	6.352
db3	SNR	7.793	5.899	4.065	6.269	8.429	7.025	5.561	5.508	7.641	2.916	6.111
db4	SNR	5.418	4.108	1.54	6.097	7.798	8.128	5.993	6.32	9.133	3.503	5.804
db5	SNR	7.084	6.759	4.234	7.769	9.044	5.137	5.42	6.237	8.74	1.608	6.203
db6	SNR	8.456	7.318	5.181	7.642	10.055	9.797	6.708	6.642	8.726	5.328	7.585
coif2	SNR	7.395	6.276	3.849	6.517	9.296	7.721	5.785	5.809	8.411	2.588	6.365
coif3	SNR	7.35	6.568	3.912	6.941	9.376	7.988	6.487	6.341	8.641	4.028	6.763
coif4	SNR	7.901	6.76	4.608	6.616	8.638	8.297	6.688	5.885	7.815	4.568	6.778
coif5	SNR	7.62	6.573	3.96	8.58	11.087	8.919	6.648	6.714	9.161	3.347	7.261
Haar	SNR	9.29	7.494	5.251	7.526	9.444	9.771	7.469	6.417	8.785	5.529	7.698

Table 11

Ablation experiments with 5 dB noise intensities under MA.

Record		103	105	111	116	122	205	213	219	223	230	Average
modelNet1	RMSE	0.188	0.285	0.213	0.514	0.34	0.183	0.41	0.385	0.26	0.306	0.308
db2	RMSE	0.223	0.295	0.247	0.448	0.281	0.195	0.411	0.364	0.244	0.339	0.305
db3	RMSE	0.187	0.275	0.217	0.502	0.309	0.176	0.393	0.367	0.252	0.295	0.297
db4	RMSE	0.23	0.301	0.251	0.464	0.283	0.209	0.426	0.363	0.244	0.343	0.311
db5	RMSE	0.206	0.28	0.224	0.478	0.299	0.16	0.387	0.363	0.245	0.31	0.295
db6	RMSE	0.195	0.279	0.214	0.494	0.319	0.147	0.37	0.364	0.234	0.293	0.291
coif2	RMSE	0.219	0.298	0.244	0.507	0.32	0.192	0.405	0.384	0.248	0.328	0.314
coif3	RMSE	0.202	0.294	0.224	0.598	0.445	0.14	0.467	0.433	0.287	0.309	0.34
coif4	RMSE	0.194	0.289	0.216	0.522	0.366	0.135	0.391	0.377	0.256	0.279	0.303
coif5	RMSE	0.195	0.285	0.213	0.5	0.33	0.138	0.369	0.362	0.232	0.28	0.29
Haar	RMSE	0.207	0.284	0.234	0.444	0.256	0.172	0.374	0.347	0.221	0.309	0.285
modelNet1	SNR	8.384	6.811	4.009	7.635	9.534	6.37	6.005	7.281	9.358	2.709	6.81
db2	SNR	8.704	6.652	3.091	9.462	11.079	6.888	6.211	8.059	9.948	2.29	7.238
db3	SNR	8.907	7.447	3.859	7.985	10.363	6.765	6.357	7.952	9.57	3.132	7.234
db4	SNR	8.199	6.521	2.971	8.974	10.917	6.23	5.854	8.09	9.918	2.392	7.007
db5	SNR	8.866	7.507	3.841	8.228	10.431	9.213	6.698	8.149	9.929	3.116	7.598
db6	SNR	7.994	6.473	3.608	7.994	9.7	9.703	6.926	8.238	10.589	3.429	7.465
coif2	SNR	7.966	6.349	2.91	7.546	9.638	6.689	6.024	7.292	9.838	2.602	6.685
coif3	SNR	7.266	5.893	2.737	6.382	6.591	9.997	5.082	5.903	8.189	2.693	6.073
coif4	SNR	7.065	6.067	2.92	7.51	8.475	9.475	6.451	7.974	9.825	3.487	6.925
coif5	SNR	7.429	5.782	3.286	7.937	9.381	10.253	6.906	8.328	10.641	3.733	7.368
Haar	SNR	8.87	7.1	3.503	9.805	12.357	8.978	7.057	8.741	10.945	3.065	8.042

DAN, FCN has a certain filtering ability, but its effect is significantly inferior to the DW-CNN. DW-CNN can effectively extract deep features and retain all effective information in the pooling process, which helps the model to make full use of the input signals. Finally, from Tables 12–20, the proposed DW-CNN achieves lower RMSE and higher SNR, which means DW-CNN has better denoise performance.

Fig. 7 shows the effect of three filtering methods under BW noise. Among them, Fig. 7 (a) shows the ECG signal with BW, Fig. 7 (b) shows the corresponding clean ECG signal, and Fig. 7 (c) to Fig. 7 (e) show the ECG signal denoised by DNN-based DAN, FCN and DW-CNN. Fig. 8 shows the effect of three filtering methods under EM noise. Among them, Fig. 8 (a) shows the ECG signal with EM, Fig. 8 (b) shows the corresponding clean ECG signal, and Fig. 8 (c) to Fig. 8 (e) show the ECG signal denoised by DNN-based DAN, FCN and DW-CNN. Fig. 9 shows the effect of four filtering methods under MA noise. Among them, Fig. 9 (a) shows the ECG signal with MA, Fig. 9 (b) shows the corresponding clean ECG signal, and Fig. 9 (c) to Fig. 9 (e) show the ECG signal denoised by DNN-based DAN, FCN and DW-CNN. As can be seen from Fig. 7, Fig. 8 and Fig. 9, it is difficult for DNN-based DAN and FCN to learn effective

network weights in the data training process, resulting in the model being unable to effectively denoise noisy signals. Finally, compared with the other existing filtering methods, DW-CNN can obtain better denoising performance under various kinds of noise, which means the filtered signal can enter the next stage of analysis in clinical application.

To sum up, the results of comparative experiments under different noise environments and noise intensities show that compared with other advanced denoising models, DW-CNN has superior denoising performance and has a certain clinical application value.

4.3. Limitations and future works

Through detailed ablation experiments and comparison experiments, the proposed DW-CNN can effectively extract deep features and filter noise from input signals. However, due to the limited number of patients, we will continue to obtain more ECG samples from practical applications in the future to further train the proposed model. In future work, we will deploy the proposed DW-CNN to practical applications to provide clean signals for subsequent diagnosis. Meanwhile, we will learn

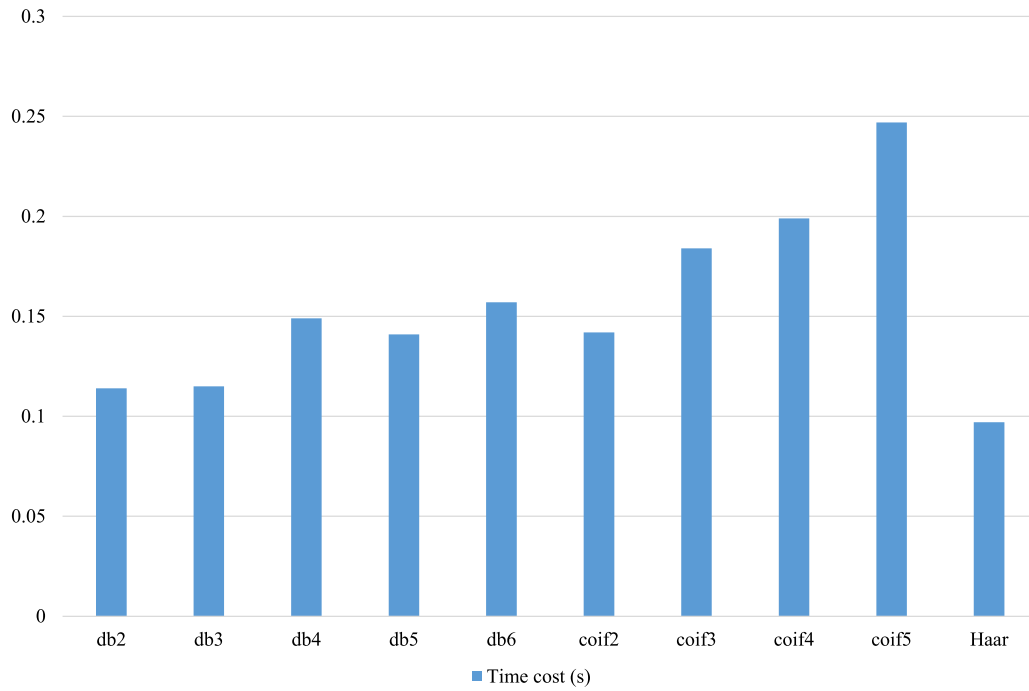


Fig. 6. The time cost of different wavelet types.

Table 12

Comparison experiments with 0 dB noise intensities under BW.

Record		103	105	111	116	122	205	213	219	223	230	Average
DAN	RMSE	0.333	0.465	0.353	0.88	0.573	0.24	0.629	0.588	0.427	0.402	0.489
FCN	RMSE	0.349	0.46	0.353	0.854	0.568	0.238	0.636	0.576	0.425	0.411	0.487
Haar	RMSE	0.165	0.324	0.254	0.646	0.291	0.16	0.224	0.398	0.195	0.28	0.294
DAN	SNR	1.034	0.305	-1.39	2.379	4.262	4.298	0.394	2.787	3.815	-0.99	1.69
FCN	SNR	0.564	0.451	-1.49	2.655	4.245	4.398	0.294	2.971	3.806	-1.22	1.667
Haar	SNR	10.186	5.868	2.351	7.691	11.345	8.36	10.657	7.122	11.306	2.24	7.713

Table 13

Comparison experiments with 1.25 dB noise intensities under BW.

Record		103	105	111	116	122	205	213	219	223	230	Average
DAN	RMSE	0.369	0.463	0.36	0.909	0.541	0.239	0.701	0.579	0.406	0.448	0.501
FCN	RMSE	0.361	0.457	0.356	0.904	0.555	0.236	0.709	0.577	0.406	0.453	0.501
Haar	RMSE	0.126	0.299	0.21	0.753	0.315	0.131	0.291	0.401	0.225	0.24	0.299
DAN	SNR	0.412	0.355	-2.058	2.771	4.484	4.353	0.25	2.8	4.3	-0.35	1.732
FCN	SNR	0.563	0.523	-1.968	2.855	4.232	4.485	0.158	2.837	4.286	-0.51	1.746
Haar	SNR	11.156	7.948	4.301	6.716	10.252	9.543	8.545	6.777	10.077	6.093	8.141

Table 14

Comparison experiments with 5 dB noise intensities under BW.

Record		103	105	111	116	122	205	213	219	223	230	Average
DAN	RMSE	0.373	0.456	0.344	0.739	0.54	0.229	0.7	0.556	0.42	0.45	0.481
FCN	RMSE	0.369	0.453	0.341	0.709	0.549	0.225	0.695	0.56	0.423	0.445	0.477
Haar	RMSE	0.221	0.316	0.225	0.428	0.238	0.203	0.29	0.327	0.179	0.319	0.275
DAN	SNR	0.584	0.071	-1.689	3.207	4.54	4.545	0.131	3.198	4.198	-1.35	1.744
FCN	SNR	0.685	0.18	-1.638	3.433	4.405	4.648	0.18	3.142	4.159	-1.21	1.799
Haar	SNR	7.282	4.907	3.467	8.783	12.136	7.593	9.227	8.597	12.769	2.132	7.689

from the application of machine learning methods in other medical fields [50–52], and further explore the application of artificial intelligence methods [53–56] in the field of ECG.

5. Conclusion

This paper proposes a novel ECG denoising method, called DW-CNN, including input layer, convolution layer, DWT-based pooling layers, and IDWT-based up-sampling layers. the convolution layer is used to replace the simple full-connected layer to construct the encoder and decoder of

Table 15

Comparison experiments with 0 dB noise intensities under EM.

Record		103	105	111	116	122	205	213	219	223	230	Average
DAN	RMSE	0.398	0.377	0.306	0.756	0.585	0.21	0.687	0.604	0.433	0.438	0.479
FCN	RMSE	0.41	0.392	0.333	0.759	0.564	0.219	0.694	0.595	0.429	0.423	0.482
Haar	RMSE	0.293	0.259	0.217	0.388	0.347	0.157	0.378	0.442	0.293	0.34	0.311
DAN	SNR	0.829	0.502	−0.57	3.271	3.965	4.929	0.02	2.66	3.779	−0.72	1.866
FCN	SNR	0.564	0.184	−1.41	3.234	4.248	4.563	−0.062	2.777	3.849	−0.60	1.734
Haar	SNR	6.745	5.314	4.182	10.511	10.507	8.895	7.294	6.442	8.541	2.176	7.061

Table 16

Comparison experiments with 1.25 dB noise intensities under EM.

Record		103	105	111	116	122	205	213	219	223	230	Average
DAN	RMSE	0.387	0.43	0.32	0.822	0.59	0.214	0.726	0.608	0.443	0.42	0.496
FCN	RMSE	0.385	0.436	0.329	0.821	0.582	0.217	0.729	0.608	0.438	0.412	0.496
Haar	RMSE	0.262	0.279	0.189	0.51	0.362	0.133	0.333	0.46	0.266	0.272	0.307
DAN	SNR	1.279	0.392	−1.037	2.759	4.019	4.773	0.169	2.732	3.764	−0.55	1.83
FCN	SNR	1.316	0.26	−1.315	2.788	4.129	4.646	0.138	2.731	3.844	−0.43	1.811
Haar	SNR	7.266	7.518	5.006	8.575	9.706	9.524	8.048	6.393	9.528	3.857	7.542

Table 17

Comparison experiments with 5 dB noise intensities under EM.

Record		103	105	111	116	122	205	213	219	223	230	Average
DAN	RMSE	0.362	0.441	0.337	0.955	0.571	0.223	0.723	0.573	0.436	0.449	0.507
FCN	RMSE	0.365	0.443	0.341	0.949	0.569	0.224	0.725	0.571	0.434	0.455	0.508
Haar	RMSE	0.196	0.277	0.217	0.686	0.271	0.155	0.346	0.354	0.229	0.313	0.304
DAN	SNR	0.531	0.145	−1.517	2.911	4.289	4.415	0.244	3.05	3.889	−1.56	1.64
FCN	SNR	0.44	0.086	−1.638	2.954	4.323	4.37	0.214	3.074	3.93	−1.65	1.61
Haar	SNR	10.127	7.607	3.778	8.609	12.158	9.141	7.588	8.473	10.95	2.184	8.061

Table 18

Comparison experiments with 0 dB noise intensities under MA.

Record		103	105	111	116	122	205	213	219	223	230	Average
DAN	RMSE	0.356	0.451	0.306	0.805	0.573	0.216	0.69	0.562	0.443	0.427	0.483
FCN	RMSE	0.359	0.438	0.333	0.83	0.559	0.219	0.686	0.554	0.431	0.426	0.484
Haar	RMSE	0.199	0.288	0.223	0.562	0.308	0.165	0.302	0.361	0.23	0.283	0.292
DAN	SNR	0.874	0.346	−0.528	2.667	4.091	4.726	0.114	3.077	3.521	−0.69	1.82
FCN	SNR	0.803	0.207	−1.414	2.39	4.262	4.563	0.174	3.18	3.746	−0.73	1.718
Haar	SNR	8.54	7.088	4.11	7.864	11.471	8.231	8.594	7.909	10.247	3.324	7.738

Table 19

Comparison experiments with 1.25 dB noise intensities under MA.

Record		103	105	111	116	122	205	213	219	223	230	Average
DAN	RMSE	0.347	0.447	0.32	0.833	0.55	0.214	0.687	0.59	0.429	0.434	0.485
FCN	RMSE	0.35	0.436	0.329	0.837	0.57	0.217	0.686	0.592	0.429	0.436	0.488
Haar	RMSE	0.166	0.276	0.178	0.562	0.378	0.127	0.354	0.433	0.27	0.251	0.3
DAN	SNR	1.135	0.049	−1.001	2.805	4.489	4.771	0.381	2.976	3.823	−0.63	1.88
FCN	SNR	1.027	0.26	−1.315	2.763	4.177	4.646	0.413	2.973	3.824	−0.69	1.807
Haar	SNR	9.29	7.494	5.251	7.526	9.444	9.771	7.469	6.417	8.785	5.529	7.698

Table 20

Comparison experiments with 5 dB noise intensities under MA.

Record		103	105	111	116	122	205	213	219	223	230	Average
DAN	RMSE	0.358	0.44	0.336	0.783	0.574	0.223	0.827	0.574	0.435	0.415	0.496
FCN	RMSE	0.365	0.443	0.341	0.805	0.569	0.224	0.748	0.571	0.434	0.428	0.493
Haar	RMSE	0.207	0.284	0.234	0.444	0.256	0.172	0.374	0.347	0.221	0.309	0.285
DAN	SNR	0.636	0.153	−1.481	3.593	4.262	4.439	−0.471	3.028	3.879	−0.45	1.759
FCN	SNR	0.44	0.086	−1.638	3.042	4.323	4.37	0.107	3.074	3.93	−0.76	1.698
Haar	SNR	8.87	7.1	3.503	9.805	12.357	8.978	7.057	8.741	10.945	3.065	8.042

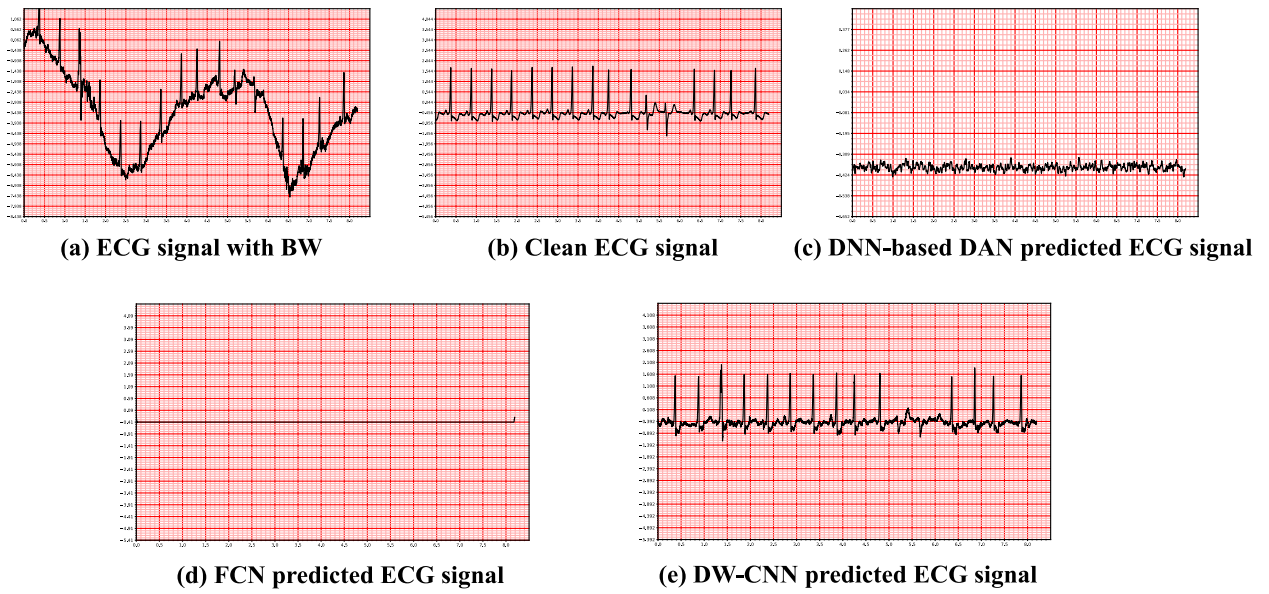


Fig. 7. The effect of three filtering methods under BW noise.

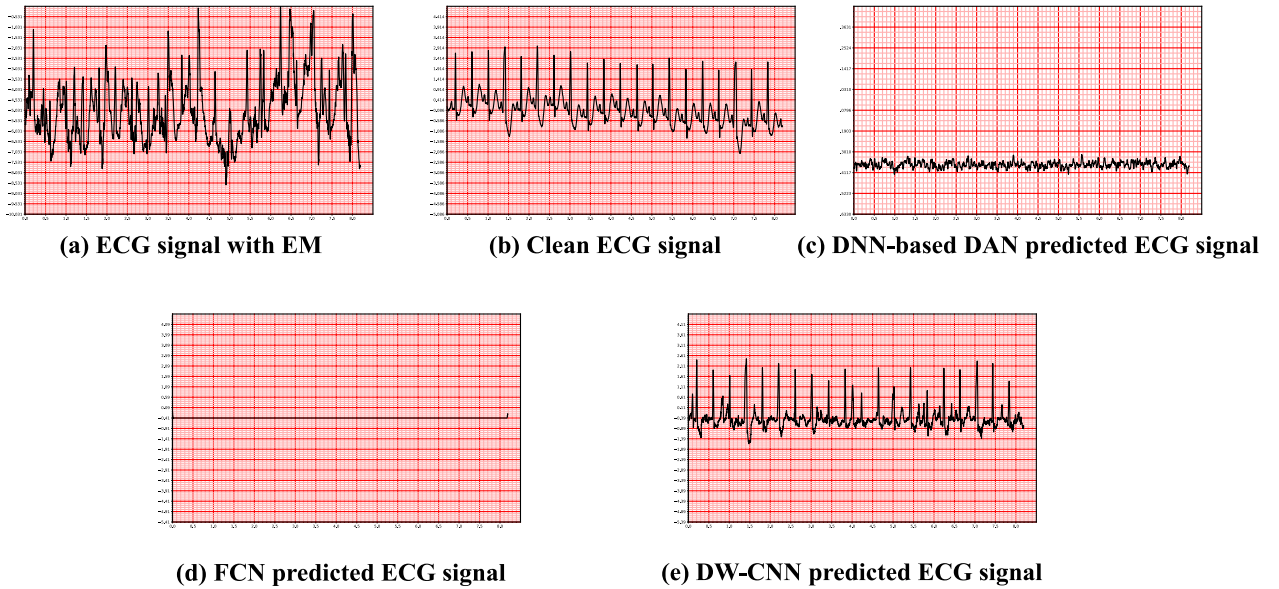


Fig. 8. The effect of three filtering methods under EM noise.

ECG signals. To overcome the problem of effective information loss caused by simply discarding information in the process of signal down-sampling caused by the pooling layer, DWT is used to down-sample the signal instead of the maximum pooling operation, to compress the spatial size of the input signal and effectively retain all the information in the input signal. Compared with the existing methods, DW-CNN has better denoising performance under different noise intensities and noise types, with an average decrease of 0.194 RMSE and an increase of 5.99 SNR.

The proposed DW-CNN can effectively extract and retain the deep features and overcome the performance limitation of deep learning method in the denoising process. In practical application of wearable devices, this method can provide clean ECG signals for subsequent diagnosis and become an important part of accurate ECG analysis system.

CRediT authorship contribution statement

Yanrui Jin: Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing, Resources, Data curation, Visualization, Validation. **Chengjin Qin:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Jinlei Liu:** Visualization, Validation. **Yunqing Liu:** Visualization, Software. **Zhiyuan Li:** Conceptualization, Methodology. **Chengliang Liu:** Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

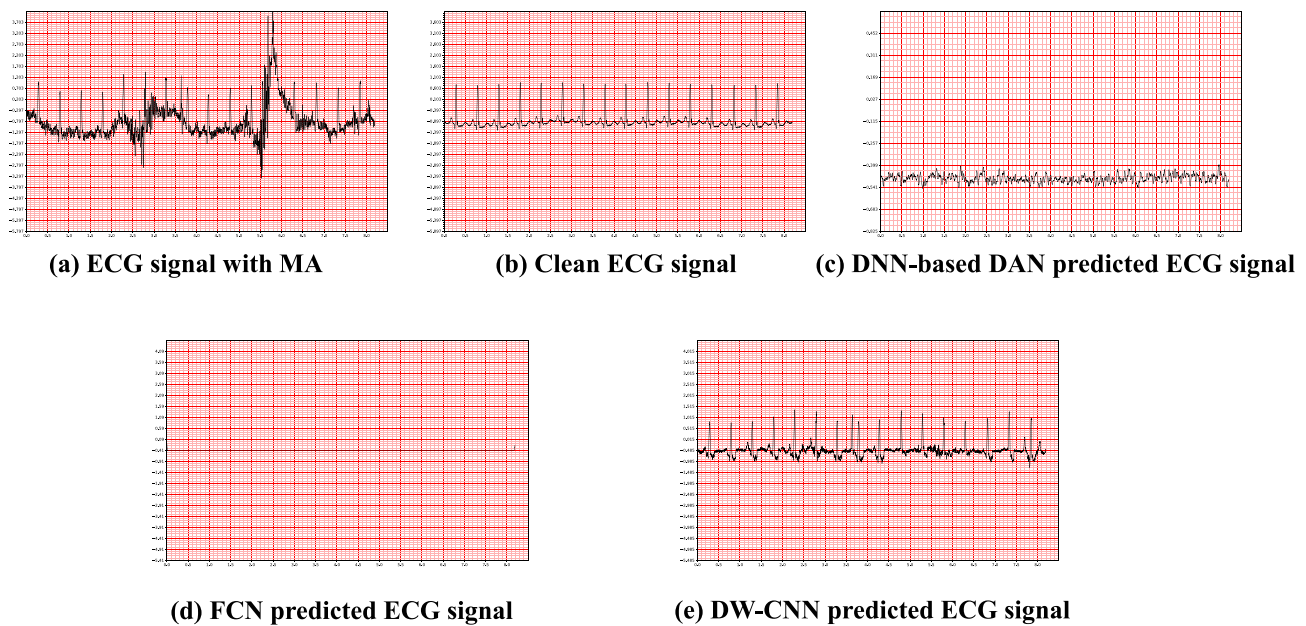


Fig. 9. The effect of three filtering methods under MA noise.

Data availability

Data will be made available on request.

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